Peterson Guo

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Education

University of Waterloo — Waterloo, ON
BASc. in Honours Electrical Engineering — *09/2023 - 04/2028*

Skills

Languages: C/C++, Python, MLIR, Bash
Software: LLVM, CUDA, ROCm, Pytorch, Git
Others: Compilers, GPU Drivers, Linux, Operating Systems, Kernel Debugging, LLMs

Experience

Deep Learning Library Performance Software Engineer Intern — 05/2026 – 09/2026
Nvidia — *Santa Clara, CA*

- Returning

Model Performance Engineer Intern — 01/2026 – 5/2026
Baseten — *San Francisco, CA*

- Optimized OpenAI Whisper inference on H100, achieving **5x** throughput and **60%** latency reduction via encoder batching, kernel fusion, and custom runtime scheduling on TRT-LLM
- Re-architected LLM inference stack in TRT-LLM, implementing custom **CUDA** kernels using CUTLASS and CuTe DSL (TMA loads, WGMMA/UMMA, swizzling, ping-pong scheduling), plus KV-cache quantization (TurboQuant) and RadixMLP broadcast/collect kernels, improving memory efficiency and long-context decode performance
- Re-architected speaker diarization pipeline, delivering **15x** throughput and **80%** latency reduction through parallelization and model-level optimizations for a **$5M** enterprise deployment
- Built a privacy-preserving LLM request hashing + replay system for deterministic workload re-simulation, enabling performance regression testing and capacity planning; presented kernel/runtime optimizations to FDEs and led technical reviews

Systems Software Engineer Intern — 05/2025 – 12/2025
Nvidia — *Toronto, ON*

- Built and owned an internal **Python–C++** profiler for CUTLASS latency decomposition with up to **21x** lower overhead than torch.profiler and adopted by **15+** systems developers
- Reduced end-to-end **JIT** compilation time by **20%** by designing a deterministic Merkle-tree hashing system **5**× faster than SHA256 for multi-GB binaries, improving kernel generation **latency** and iteration speed
- Reduced kernel launch overhead by **60%** by eliminating **Python→C++→CUDA** dispatch hotspots, achieving performance within **10%** of tvm-ffi and over **2x** faster than Triton
- Core contributor to an **MLIR**-based CUDA dialect, designing and lowering **20+** ops to enable Python DSL access to graphs, streams, and async memory operations

Software Engineer Intern — 09/2024 – 12/2024
AMD — *Markham, ON*

- Delivered AMD's most stable GPU software release, by resolving **25+** kernel-level crashes, hangs, and performance regressions, using crash dumps, **firmware** logs, and **register-level** analysis
- Shaped display-pipeline research, shown by my **ML upscaling** PoC being adopted and later expanded into an internal technical conference paper by my successor, by building a lightweight **PyTorch/ROCm** prototype
- Strengthened display-pipeline stability on next-gen AMD GPUs/APUs by building **C/C++ kernel** drivers, ensuring robust framebuffer→display behavior for Ryzen AI and Radeon hardware
- Enhanced **Linux** display pipeline correctness, by resolving defects in color calibration, frame-sync logic, DSC decoding, and corruption, contributing patches to AMD's **open-source** Linux display driver

Security Developer Co-op — 01/2024 – 5/2024
eSentire — *Waterloo, ON*

- Performed **LLM security research** on production backends, analyzing attack surfaces including prompt injection, data exfiltration, and privilege escalation in deployed pipelines
- Cut **ingestion latency** by **50%+** by **optimizing** JSON parsing and Snowflake execution paths, reducing pipeline stalls in **real-time analytics**
- Improved log-processing throughput by **4**× through **algorithmic optimizations** in **Python**'s data pipeline
- Built an AI threat analytics dashboard end-to-end (Snowflake, **Python**, Vue3) used by enterprise clients to investigate live security events, earning executive visibility

Projects

Chess Bot | PyTorch, Python, C++, CUDA, Alpha Beta Pruning, NNUE

- Built a **GPU-accelerated** chess engine using custom **CUDA** kernels for evaluation/search, achieving major speedups over CPU baselines

InvestIQ | Python, C++, IBKR API, CUDA

- Built a low-latency, event driven market data pipeline aggregating price, event, and sentiment feeds, enabling deterministic backtesting and feature generation for pairs trading and signal research.

ML Upscaling | ROCm, CUDA, Machine Learning, CNNs, PyTorch, Python

- Prototyped a real-time **ML** upscaling model for **AMD graphics cards**, leveraging **transformer** based **super sampling** to enhance visual fidelity and performance for potential display pipeline adoption